

Derbyshire Shared Care Pathology Guidelines

Polycythaemia

Signs & symptoms of polycythaemia

- May present with arterial or venous vascular occlusive events (due to increased blood viscosity)
- Occasionally haemorrhage, splenic pain, pruritus, gout, tiredness
- Many are asymptomatic and incidentally found to have abnormal full blood count

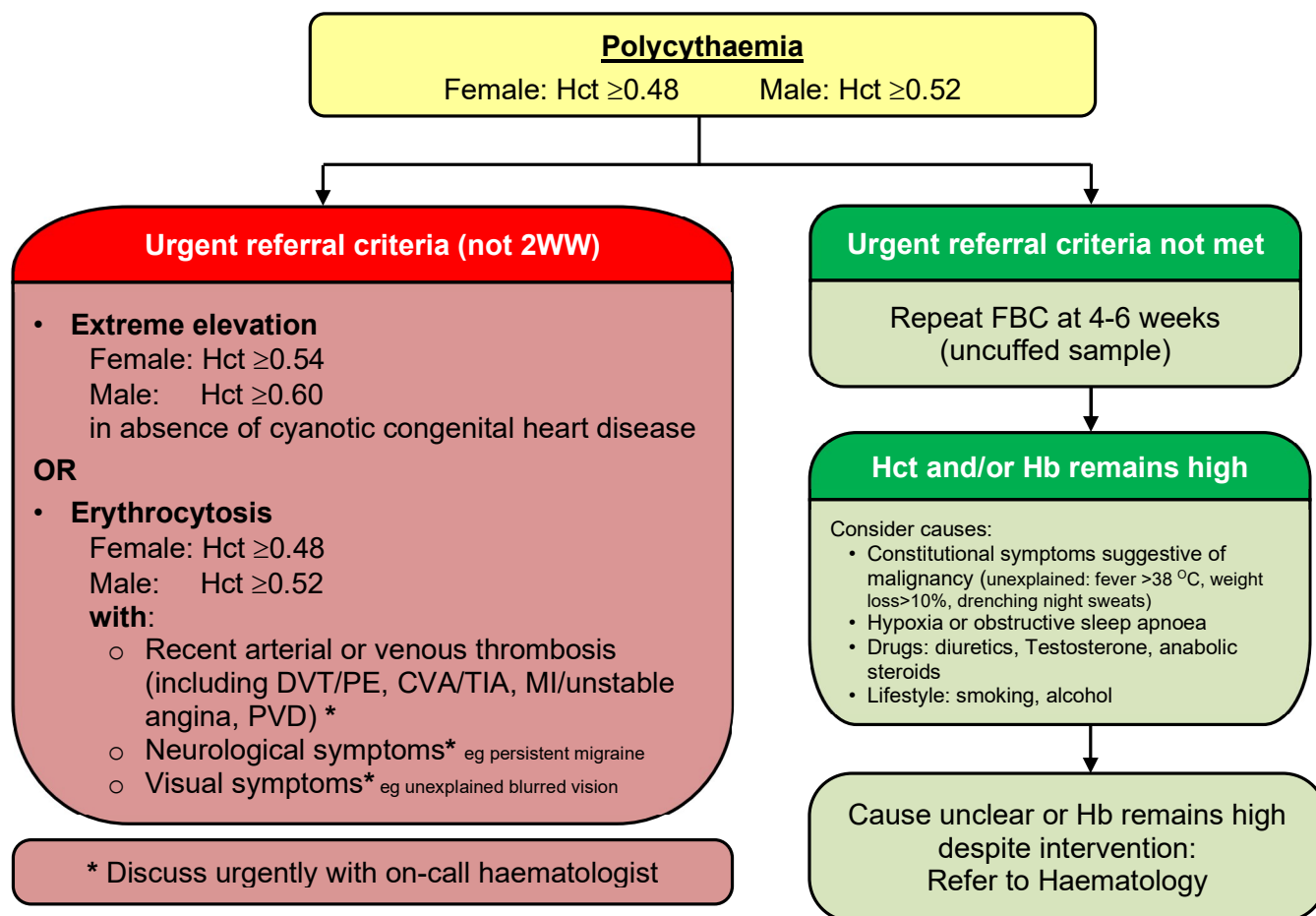


The laboratory usually telephones raised haematocrit results to GP practices (when open) when the level is >0.55 unless the patient is a known Haematology patient.

What are the main causes of polycythaemia?

A raised haemoglobin/ haematocrit is a common finding. The cause may be:

- Primary polycythaemia - myeloproliferative disorder/ polycythaemia rubra vera
- Secondary polycythaemia - secondary to other conditions, e.g. hypoxia, renal or hepatic cancer, uterine fibroids
- Apparent polycythaemia - reduction in the plasma volume is usually related to modifiable factors e.g. raised BMI, heavy smoking, excess alcohol, diuretics use



Note: If your patient has low haematinics, please do not initiate supplementation before haematology review

Polycythaemia Rubra Vera (PRV)

PRV is a primary myeloproliferative disorder and should be differentiated from those with apparent polycythaemia.

PRV is defined as an acquired increase in haemoglobin/haematocrit (Hct) level above 165 g/L/0.49 in men and 160g/L/0.48 in women, in the context of a *JAK2* mutation and characteristic bone marrow morphology.

JAK2 molecular test will be requested at haematology clinic to ascertain the diagnosis.

This guideline is to help clinicians in identifying those patients requiring further investigations. It does not replace advice and guidance or letter correspondence requests to a haematology consultant if you are in any doubt and need discussion.

Some helpful reference information about polycythaemia that sits behind this guideline:

Is polycythaemia the same as erythrocytosis?

Polycythaemia and erythrocytosis are technically different and are often incorrectly used interchangeably. Polycythaemia literally means "lots of cells in the blood", so this can encompass more than just red cells to include white cells and platelets too. Erythrocytosis pertains specifically to more (and a higher concentration of) red cells.

The flow chart shows a repeat sample should be 'uncuffed'. Why is this needed?

A cuffed sample (using a tourniquet as is normal practice when taking bloods) would in theory mean the blood is stagnant and this can make the sample more concentrated (so affect the haematocrit). The British Society for Haematology and local expert interpretation suggest a repeat uncuffed sample (no tourniquet when taking), to eliminate the circumstantial 'one-off' raised reading being due to stagnant blood.

If a raised haematocrit doesn't meet urgent referral criteria, why is it suggested to repeat the FBC in 4-6 weeks' time? Is this timing critical?

Repeating in 4-6 weeks' time eliminates other circumstantial 'one-off' raised readings due to transient problems e.g. dehydration, period of heavy drinking. Often the haematocrit would revert back to normal without intervention. However, if there is significant concern, there is no need to stick with the suggested time-frame.

When can I assume that lifestyle factors are the cause of a raised (but not significantly raised) haematocrit and why do they contribute?

We can't be prescriptive about every scenario and so it will depend on the clinical context (hence using dialogue with a consultant haematologist if unsure). Apparent erythrocytosis is increased haematocrit and haemoglobin concentration with a normal red-cell mass. It is

caused by a low plasma volume that most commonly occurs in people taking diuretics, and those with high alcohol intake. Other lifestyle factors can cause a raised haematocrit through mild but persistent hypoxia, and may be modifiable e.g. reducing smoking, obesity.

References

1. NICE CKS <https://cks.nice.org.uk/topics/polycythaemia-erythrocytosis/>
2. McMullin, Mary Frances, et al. "A guideline for the diagnosis and management of polycythaemia vera." British journal of haematology 184.2 (2019): 176-191.

Contacts

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Document title	Polycythaemia		
Revision	3	Document Reference	CHISCP60
Reason for review	Routine Review		
Review history	Version/Date	Authors	Reason for review
July 2022	V1	Dr M Khalifa, Dr E Welch, Dr Richard Went	New Document
October 2024	V2	Dr F Al Kaisi, Dr K Lam, Dr J Tam Dr P Blackwell, Ms S Knowles	Routine review
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